Domain-Adaptive Pretraining for Statutory Question Answering: Evidence from Authentic, Synthetic, and Mixed Legal Text

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Abstract

Statutory question answering requires models to extract precise spans from structured legal text. Domain-adaptive pretraining (DAPT) adds statutory exposure, but better masked language model fit may not improve extraction. We test this with InLegalBERT and IndicIPR-QA, an English benchmark covering seven Indian intellectual property Acts. The main comparison fixes DAPT at 100 optimiser updates with 10 warmup updates. Across five seeds, we compare authentic statutory text, synthetic text generated from task questions and contexts, a mixture of the two, and no additional DAPT. Mean test F1 is 85.86 without DAPT, 85.85 with authentic text, 85.91 with synthetic text, and 85.58 with mixed text. A condition test with seed as a block detects no difference ($\boldsymbol{p = 0.706}$), and all paired 95% intervals include zero. An earlier comparison trained for three epochs gives the same general result despite unequal update counts. Masked language model diagnostics improve without a matching extraction gain. Comparisons using 25% and 50% of the answer labels do not establish that adaptation becomes more beneficial as labelled data decrease. Under the tested benchmark, encoder, corpus construction, and short adaptation budget, DAPT does not reliably improve extraction. Legal DAPT evaluations should report language model fit and task performance separately, alongside update counts, corpus provenance, and seed uncertainty.

Keywords: Legal NLP, Domain-adaptive pretraining, Extractive question answering, Synthetic text, Statutory language, Indian intellectual property law

1 Introduction

Legal extractive question answering (QA) requires a reader to identify an answer span in a source paragraph. Statutes contain enumerated clauses, conditions, exceptions, and institutional terminology. A model may recognise this language yet select a span that omits a qualification or extends into an adjacent clause. Additional exposure to statutory text is therefore a plausible adaptation strategy, but improved language model fit need not improve the final answer.

Domain-adaptive pretraining (DAPT) continues a pretrained model's self-supervised training on domain text (Gururangan et al. 2020). Legal domain encoders already provide substantial legal language exposure (Chalkidis et al. 2020; Paul et al. 2023). Here, DAPT is an umbrella term for the masked language model continuation stage. The synthetic and mixed routes use continued pretraining informed by the task. They are not adaptation based on domain text alone because their corpora use benchmark questions and contexts, although they exclude answer strings and offsets. The question here is narrower than whether legal pretraining helps a generic encoder: does additional statutory adaptation targeted to the benchmark improve an InLegalBERT reader after the same supervised transfer and fine-tuning stages?

The choice of adaptation text complicates this comparison. Authentic provisions retain statutory wording, whereas generated passages can extend the adaptation pool without adding answer labels. Their combination may change vocabulary, drafting patterns, and repetition. Corpus size, token exposure, optimisation, and checkpoint selection can also differ. A high score from one adapted checkpoint cannot identify which factor produced it.

The analysis uses IndicIPR-QA, a benchmark of 4,462 question–answer pairs over seven Indian intellectual property Acts. Its construction and validation are described in a separate accepted resource paper (Anonymous 2026). The present study holds the annotated source paragraph available and studies reader adaptation; provision retrieval is outside its scope. The evidence contains 62 independently trained readers. It includes a comparison with equal update counts on the full dataset, an earlier comparison trained for three epochs, a control written in statutory style, and controls for the fraction of answer labels.

The study addresses four research questions:

RQ1: Efficacy.
: Under the executed training recipe, does additional statutory DAPT improve the otherwise fixed InLegalBERT–SQuAD–IndicIPR-QA route?

RQ2: Composition.
: Under an MLM schedule with equal update counts, do authentic, synthetic, and mixed adaptation corpora yield distinguishable answer extraction performance on the full dataset across training seeds?

RQ3: Target alignment.
: Does synthetic text seeded by the target task outperform an untargeted synthetic control written in statutory style?

RQ4: Answer label fraction.
Does the DAPT effect change when final IndicIPR-QA fine-tuning uses answer labels from fewer training sections?

The contribution is an empirical audit rather than a new encoder architecture. First, a post hoc sensitivity comparison across five seeds matches MLM update counts, warmup, and the learning rate schedule across authentic, synthetic, and mixed DAPT. Second, a completed study with three training epochs, provenance checks, and token budget audits show which conclusions survive under a different exposure rule. Third, paired controls for writing style and the fraction of answer labels test narrower explanations for any apparent gain. Corpus, answer difficulty, and masked language model (MLM) diagnostics support interpretation without substituting for downstream QA evaluation.

This distinction is the main reason the absence of a detected effect is informative. Validation MLM scores are often used to show that an encoder has adapted to a domain, but improved reconstruction of statutory tokens is not itself evidence of improved span extraction conditioned on the question. Evaluating both objectives under the same downstream route tests whether the apparent language model adaptation transfers to the task for which it is intended.

The test with equal update counts on the full dataset does not detect a condition effect. The earlier test with three training epochs reaches the same general finding. Synthetic DAPT gives a small positive contrast when fine-tuning uses 50% of the answer labels, but neither the interaction with the fraction of answer labels nor the target alignment contrast establishes a broader advantage. These are bounded findings: the adaptation budget is small, the conditions with equal update counts process different numbers of nonpadding tokens, and only one corpus sample is used per condition. The study therefore distinguishes reproducible scores from stronger causal or legal conclusions.

The following sections situate the work, define the benchmark and executed protocol, and report the completed results and diagnostics. The discussion addresses legal interpretation and limits. Archived experiments and validation only QA summaries are retained in the appendices rather than used as substitutes for the final test comparisons.

2 Related work

2.1 Domain-adaptive and synthetic continued pretraining

BERT introduced bidirectional masked language model pretraining and showed that the resulting encoder could be fine-tuned for extractive QA and language inference with only small task specific output layers (Devlin et al. 2019). Gururangan et al. then evaluated a second pretraining phase on four domains and eight classification tasks. They reported gains from domain-adaptive pretraining in both high- and low resource settings, with further gains from adapting to unlabeled task data (Gururangan et al. 2020). Related domain specific models use different training routes. SciBERT is pretrained on a corpus of scientific publications, whereas BioBERT starts from BERT

and continues pretraining on PubMed abstracts and PubMed Central articles (Beltagy et al. 2019; Lee et al. 2020).

Synthetic continued pretraining offers another route when only a small source corpus is available. EntiGraph extracts salient entities from that corpus and generates new text by connecting sampled entities (Yang et al. 2025). TOP Training extracts entities from a target medical extractive QA dataset, uses an LLM to generate texts that contain those entities, and pretrains on the generated text (Sengupta et al. 2025). TOP Training is a procedural precedent here, not the name of the present method. TRAIT combines in domain data selection with task oriented synthetic passage generation for continued pretraining, but evaluates advertising and mathematics rather than law (Liang et al. 2024). This legal study uses rules and keyphrases to guide unlabeled statutory style passages, then asks a different question: under a specified downstream route and adaptation budget, are authentic corpora, synthetic corpora informed by the task, and mixed corpora associated with different results?

Kang et al. compare natural web text, rephrased text, generated textbooks, and natural–synthetic mixtures across model sizes and data budgets. Their results show that the effect of synthetic data depends on its type, mixture ratio, model size, and budget (Kang et al. 2025). In a Taiwanese legal study, continual pretraining improved domain knowledge but did not improve every evaluated legal task (Chen et al. 2025). In a separate medical study, seven adapted language models and two adapted vision–language models were compared directly with their base models. Nearly all failed to improve consistently in zero- and few-shot medical QA (Jeong et al. 2024). These task specific results caution against treating domain adaptation as an automatic downstream gain. They motivate the present separation of MLM fit, fixed checkpoint QA performance, and training seed variation.

2.2 Legal language models and legal QA

Greco and Tagarelli provide a systematic review of Transformer-based language models across legal tasks and applications (Greco and Tagarelli 2024). Legal NLP has used both broad legal domain encoders and jurisdiction specific models. LEGAL-BERT compares using BERT directly, continuing its pretraining on legal corpora, and pretraining a legal model from scratch (Chalkidis et al. 2020). Paul et al. continue the pretraining of LegalBERT and CaseLawBERT on Indian legal text and also train a custom vocabulary model from scratch (Paul et al. 2023). LeXFiles is a multinational English legal corpus, while MultiLegalPile contains 689 GB of legal text from 17 jurisdictions in 24 languages (Chalkidis et al. 2023; Niklaus et al. 2024). The Pile of Law provides about 256 GB of open source English legal and administrative text and treats responsible filtering as part of corpus construction (Henderson et al. 2022). Nair and Modani examine another design choice: adding objectives that model two characteristics of legal language alongside ordinary language model pretraining (Nair and Modani 2023).

The value of legal domain and jurisdiction specific text is also task dependent. Zheng et al. compare general and legal domain pretraining on three legal tasks, including CaseHOLD. Their results associate larger gains with greater task domain specificity and closer alignment between the pretraining corpus and downstream task

(Zheng et al. 2021). On the AusLaw citation benchmark, a 7B model pretrained only on 0.5 billion Australian legal tokens slightly outperformed a model with the same backbone that had been pretrained on 94 billion mixed jurisdiction legal tokens, after both received the same citation instruction tuning. The authors present this comparison as a preliminary experiment, so it supports a task specific result rather than a general claim about legal pretraining (Han et al. 2026).

Legal benchmarks also cover different tasks. LexGLUE combines several English legal language understanding datasets in one evaluation suite (Chalkidis et al. 2022). CUAD instead contains more than 13,000 expert annotations for identifying contract text that may require human review (Hendrycks et al. 2021). These settings differ from extractive QA over statutes.

Recent work in *Artificial Intelligence and Law* covers several related but distinct QA settings. Yuan et al. use GPT-3 to generate questions for a public legal question bank and compare machine generated with human written questions (Yuan et al. 2024). DiscoLQA evaluates discourse based, zero-shot answer retrieval over six European norms (Sovrano et al. 2025). Italiani et al. generate synthetic legal context–question–answer triples with a frozen LLM and use them to train smaller supervised models in a retrieval based setting (Italiani et al. 2025). AILQA evaluates retrieval augmented QA for Indian legal material with automatic metrics and expert legal feedback, and also reports an All India Bar Examination evaluation (Nigam et al. 2026). These papers address different data, outputs, and evaluation protocols. The present paper focuses more narrowly on how reader pretraining affects extractive QA when the annotated statutory paragraph is supplied.

2.3 Synthetic text as adaptation material

Italiani et al. identify the cost of carefully annotated legal QA data as a barrier and use a frozen LLM to generate labelled examples for supervised student model training (Italiani et al. 2025). This paper uses synthetic text differently. The generated passages are not treated as additional gold QA labels and are not used to teach the model a particular answer. They are used only during MLM pretraining, where the task is to reconstruct masked tokens from local context. This distinction matters because a synthetic passage may still provide useful adaptation text even when it has not been verified as a statement of law.

The distinction also creates a different failure mode. Synthetic passages may carry legal vocabulary and statutory form, but they may also carry generator artifacts, repeated templates, or simplified legal structure. A strong downstream result after synthetic DAPT would therefore not, by itself, show that the model learned authentic legal reasoning. It could reflect useful lexical coverage, regularisation, or accidental alignment with the benchmark's question style. The corpus composition question has to be measured with controls rather than inferred from one high performing route. Here, "corpus composition" denotes the combined provenance and properties of the constructed adaptation corpus, including authenticity, generation process, lexical and length distributions, and target conditioning. The design does not identify which individual property would cause an observed difference. More specifically, A versus S/M changes provenance and several correlated linguistic properties; S versus Style tests

target conditioning while holding the generator and construction procedure broadly fixed; and token matching controls approximate exposure scale. None of these comparisons independently identifies lexical distribution, statutory structure, or factual accuracy as a causal factor.

2.4 Why corpus composition claims need audit

DAPT experiments are especially vulnerable to confounding because the adaptation stage is usually cheap to vary in several ways at once. A model may receive more tokens, more duplicate passages, longer training, a different checkpoint selection rule, or a different random seed, while the resulting table labels the difference as "authentic" or "synthetic" text. This is a problem for scientific interpretation even when all individual training runs are correctly implemented. The correct comparison must specify the unit of exposure, the eligibility rule for pretraining text, the amount of text, the training budget, and the downstream pipeline. Repeated full pipeline seeds are particularly important. Mosbach et al. show that changing only the random seed can produce substantial variation when fine-tuning BERT, RoBERTa, and ALBERT on three GLUE datasets (Mosbach et al. 2021).

The statutory setting makes this audit more important. Legal provisions contain highly repeated formulae, numbered clauses, definitions, exceptions, and institutional roles. Repetition is a real property of statutes, but it can also make DAPT corpora look larger than they are if duplicate records are counted as independent adaptation examples. The present paper therefore distinguishes archived evidence matched by record count, the completed study with three epochs and approximately matched token exposure, and the later sensitivity comparison with equal update counts. The archive matched by record count does not match token exposure. The study with three epochs closely matches its initial token budget but not its update count. The comparison with equal update counts matches the schedule but processes different numbers of nonpadding tokens because the corpora have different length distributions. None warrants an unrestricted causal claim about corpus composition.

3 Task and benchmark

The task is extractive question answering with the gold statutory paragraph provided. Each input is a pair (q, c), where q is a natural language question and c is the annotated source paragraph. The model predicts start and end token positions, which the evaluator converts to a contiguous character span in c. Performance is reported as exact match (EM) and SQuAD style F1 at token level, as used in extractive QA (Rajpurkar et al. 2016). The evaluator lowercases text, removes punctuation and normalizes whitespace; articles are retained. Retrieval is outside the scope of this manuscript.

The experiments use IndicIPR-QA, an extractive benchmark over seven Indian intellectual property Acts: Patents, Trade Marks, Copyright, Designs, Geographical Indications, Semiconductor Layout-Design, and Plant Varieties. The dataset contains 4,462 question–answer pairs, split into 3,710 training, 408 validation, and 344 test items. The split is controlled at the statutory section level in the released benchmark

Table 1 Dataset facts required for the DAPT study. Benchmark construction and human validation are reported in the companion resource paper; this manuscript uses the benchmark as an evaluation instrument.

Item	Value
Legal domain	Indian intellectual property statutes
Statutory instruments	7 Acts
QA pairs	4,462
Unique curated contexts	1,116
Train / validation / test	3,710 / 408 / 344
Mean question / context / answer length	11.90 / 171.27 / 35.64 words
Evaluation setting	Gold statutory paragraph supplied
Primary metrics	EM and token F1

so that held out questions test unseen source sections rather than paraphrases of training questions from the same section. Table 1 summarises the dataset facts used by this paper.

Supplying the gold paragraph fixes upstream evidence identification and isolates the question DAPT can directly affect: answer boundary prediction within a source provision identified by the benchmark. Provision retrieval is evaluated in separate work.

The evaluation target also matches the legal use case. A reader that extracts a span from a statute must identify answer boundaries inside long, syntactically dense provisions. Errors may involve omitted conditions, extra provisos, truncated institutional roles, or confusion between a rule and an exception. Token F1 describes partial token overlap, while exact match measures equality with the normalised annotation. Neither directly measures whether an answer preserves the provision's legal effect or all necessary qualifications.

4 Executed Study Design

4.1 Conditions and Replication

All main study training/evaluation runs start from `law-ai/InLegalBERT` (Paul et al. 2023). Condition N omits additional MLM training; A uses authentic training split provisions, S uses synthetic statutory style passages generated from task questions and contexts, and M mixes authentic and synthetic passages at approximately equal token shares. The style control uses synthetic text matched by construction and written in statutory style without conditioning on the seven target Acts or their benchmark derived concepts. Table 2 summarises the executed design.

The experimental replication unit is a training seed, not an individual prediction pooled across repeated readers. Experiment D was added after the three-epoch study as a post hoc schedule sensitivity analysis. It reuses the five N readers because N has no MLM stage, but trains new A, S, and M routes. Experiment B reuses the corresponding S readers from A. Experiment C reuses the SQuAD checkpoints from A that match both condition and seed and changes only the final IPR fine-tuning

Table 2 Completed experiment inventory. D is the sensitivity analysis with equal update counts; A–C are the earlier three-epoch studies. N, A, S, M, and Style denote adaptation conditions. Reused readers are not counted twice.

Experiment	Comparison	Independent reader runs
D: equal updates	N/A/S/M; seeds 7, 42, 99, 123, 2026; N reused from A	15 additional
A: three epochs	N/A/S/M; seeds 7, 42, 99, 123, 2026	20
B: alignment	Style; seeds 7, 42, 99; paired with matching S runs from A	3 additional
C: answer labels	N/A/S/M at 25% and 50%; seeds 7, 42, 99	24 additional
Evaluation	408 validation and 344 test questions for every reader	62 unique readers
MLM diagnostic	Base encoder and the 18 A/B adapted checkpoints	19 encoders; validation only

data. The 62 unique readers require 33 MLM, 38 SQuAD, and 62 IPR training stages. Corpus selection and answer label subsets are fixed rather than resampled for each training seed.

4.2 Synthetic Text and Source Provenance

Seed extraction combines legal pattern rules with KeyBERT keyphrases (Grootendorst 2020), using context and question text rather than answer strings or answer offsets. Gemma 3 models (Gemma Team 2025) generate unlabelled statutory style passages, not supervised QA pairs. This corpus construction uses task information, although the subsequent MLM objective is self-supervised.

The raw seed artifact requires a qualification. Its 5,859 records include 881 distinct exemplar contexts, three of which belong to the test split. Those exemplars are associated with 19 seed entities. The saved propagation audit checks primary and support seeds in the generation records and finds that none of those 19 entities was sampled among the 1,282 distinct entities used for generation. Consequently, no generated passage is removed by the seed provenance quarantine. The evidence supports lack of propagation of this raw artifact defect, not a claim that the original seed file was entirely training only.

The archived accepted pool contains 13,211 distinct passages: 9,294 generated by Gemma 3:12B and 3,917 by Gemma 3:27B. The final S and M corpora sample only the audited 12B pool. Exact normalised held out context matching and a 25-token overlap audit detect no held out reproduction in the eligible synthetic pool. Shorter overlaps exist, as expected for repeated statutory formulae; the checks do not prove semantic independence or reveal the generator's original pretraining data.

The style only control uses the same 12B generator, decoding settings, eight drafting style families and shared filtering code, with manually written generic concepts and exemplars from untargeted statutory domains. It is not conditioned on

IndicIPR-QA seeds. This is a construction based control for target conditioning, not a human validated semantic alignment measure. Table 3 records the common generation settings.

Table 3 Generation settings and scope of the automated checks. The final pool seeded by the target task uses Gemma 3:12B; the two model archive is not the final experimental mixture.

Component	Executed setting
Target seed selection	Frequency ranked raw pool capped at 1,500 entries; primary/support seed provenance audited as described in the text
Style families	Definition; registration; rights/remedies; authority/procedure; conditions/compliance; penalty/enforcement; licensing/assignment; dispute/appeal
Generation backend	Local Ollama; Gemma 3:12B for final S, M and Style
Decoding	Temperature 0.40; top-$p = 0.85$; repeat penalty 1.12; context length 4,096; maximum generated length 250 tokens
Output checks	Nonempty text, 70–280 words, at least two sentences, required concept, and legal style lexical markers
Rejections	Prompt/placeholder leakage, markdown, question marks, bullet format, prohibited openings, and severe repetition
Deduplication	Exact normalised text duplicates removed from merged pool
Not established	Legal correctness, doctrinal consistency, or suitability of generated text as legal authority

4.3 Token Matching and the Executed Optimisation Budget

A contains the 898 distinct contexts in the training split. S is sampled to approximately A's token budget. M allocates approximately half of its tokens to authentic text and half to a separately selected synthetic subset. The corpus selection seed is 20260829. Token counts include special tokens after truncation to 256 tokens and exclude padding. The same fixed A, S, and M corpora are used in Experiments A and D. Table 4 reports the three-epoch exposure used in A–C.

Token totals differ by less than 0.05%, but record counts and optimiser updates are not matched. With batch size 8 and gradient accumulation 4, three epochs produce 81–87 updates. All A–C MLM runs therefore finish during the 100-update warmup rather than completing the nominal linear learning rate schedule. Different update counts also imply different integrated learning rate exposure.

Experiment D addresses this schedule difference. Each adapted route completes exactly 100 optimiser updates, uses 10 warmup updates, reaches the configured peak learning rate, and finishes the linear decay at zero. Table 5 reports the telemetry.

Table 4 Adaptation corpora and three-epoch MLM budget used in Experiments A–C. Tokens are counted per epoch. Updates cover three epochs; every condition has 100 warmup updates.

Condition	Records	Nonpadding tokens	Updates	Warmup
A	898	128,727	87	100
S	861	128,713	81	100
M	879	128,663	84	100
Style	858	128,723	81	100

Equal update counts do not give equal token exposure: the mean processed total is about 445,757 tokens for A, 477,072 for S, and 460,828 for M. The differences follow from the fixed batch construction and the corpora's different length distributions.

Table 5 Executed MLM budget with equal update counts in Experiment D. Token totals are means and ranges across five seeds and exclude padding. N has no MLM stage.

Condition	Updates	Warmup	Processed nonpadding tokens
A	100	10	445,757 (444,219–447,947)
S	100	10	477,072 (476,715–477,394)
M	100	10	460,828 (460,181–461,490)

Experiment D controls update count and learning rate trajectory, while Experiment A controls the initial corpus token budget more closely. Together they provide two bounded comparisons rather than a complete isolation of corpus composition.

4.4 Training and Checkpoint Selection

Algorithm 1 summarises the executed routes. MLM uses the terminal checkpoint after either the three-epoch A–C schedule or the 100-update D schedule, with no validation loss checkpoint selection. SQuAD transfer and IPR fine-tuning select the lowest validation loss checkpoint. Table 6 gives the executed hyperparameters. The SQuAD v1 subsets contain 20,000 training and 2,000 validation examples, sampled with seed 42 and held fixed across conditions. Training seeds are applied to the model and training data order.

Each planned route is represented by $r = (d, s, f, b)$, where d is the adaptation condition, s is the training seed, f is the percentage of IPR answer labels used for final fine-tuning, and b is the MLM schedule. The experiment plan contains only the combinations executed in Experiments A–D.

MLM masks tokens with probability 0.15. Training is CPU based without half precision. Saved stage scripts and their hashes identify the executed settings; the amended analysis record distinguishes these scripts from the unrecoverable earlier runner version (Section A).

Algorithm 1 Audited DAPT and reader evaluation procedure

Input: Experiment plan $\mathcal{P}$, audited corpora $\mathcal{C}_d$, base encoder θ_0, fixed SQuAD subsets, and fixed IPR splits

Output: Per-route validation and locked-test predictions with annotation-relative EM and F1 scores

for all routes $r = (d, s, f, b) \in \mathcal{P}$ **do**
  $\theta \leftarrow \theta_0$; set training seed to s
  if $d \neq N$ **then**
    $\theta \leftarrow \text{MLMAdapt}(\theta, \mathcal{C}_d, b)$ ▷ *three epochs or 100 updates*
    Retain the terminal MLM checkpoint ▷ *no validation selection*
  else
    Retain θ_0 ▷ *no additional DAPT*
  end if
  $\theta_{\text{SQ}} \leftarrow \text{FineTuneSQuAD}(\theta)$
  Select the lowest-validation-loss SQuAD checkpoint
  $T_f \leftarrow \text{FixedIPRSubset}(f)$ ▷ $f \in \{25, 50, 100\}$
  $\theta_{\text{IPR}} \leftarrow \text{FineTuneIPR}(\theta_{\text{SQ}}, T_f)$
  Select the lowest-validation-loss IPR checkpoint
  for all (q_i, c_i, a_i) in the validation split **do**
    $\widehat{a}_i \leftarrow \text{Decode}(\theta_{\text{IPR}}, q_i, c_i)$ ▷ *common span decoder*
    Score $\widehat{a}_i$ against annotation a_i
  end for
  Store validation results
  Verify the frozen test analysis lock ▷ *no test-based selection*
  for all (q_i, c_i, a_i) in the locked test split **do**
    $\widehat{a}_i \leftarrow \text{Decode}(\theta_{\text{IPR}}, q_i, c_i)$ ▷ *same span decoder*
    Score $\widehat{a}_i$ against annotation a_i
  end for
  Store test results separately
end for
Pair conditions by seed and compute the specified contrasts

4.5 Scope of Fine-Tuning with Fewer Answer Labels

The 25% and 50% subsets sample statutory sections, stratified by Act, using seed 20260829. They contain respectively 135 and 270 of 541 training sections, yielding 942 and 1,852 of 3,710 QA pairs (25.39% and 49.92%). Each fraction uses one fixed subset shared across conditions and training seeds. The validation and test sets are unchanged.

These fractions reduce final supervised IPR fine-tuning, not all access to the task resource. Adaptation corpora remain derived from the full eligible training resource, including question derived generation seeds, and all conditions retain SQuAD transfer. The experiment therefore tests reduced fine-tuning with answer labels while broader task informed text remains available. It does not measure a proportional reduction

Table 6 Executed training settings. Batch denotes per device batch size times gradient accumulation. D denotes the sensitivity analysis with equal update counts; A–C denote the earlier three-epoch studies.

Setting	MLM D	MLM A–C	SQuAD	IPR
Maximum tokens	256	256	384	384
Window overlap	—	—	128	128
Budget	100 updates	3 epochs	2 epochs	3 epochs
Batch	8×4	8×4	16×2	16×2
Learning rate	2×10^{-5}	2×10^{-5}	3×10^{-5}	2×10^{-5}
Weight decay	0.01	0.01	0.01	0.01
Warmup updates	10	100	200	50
Scheduler	Linear	Linear	Linear	Linear
Checkpoint	Terminal	Terminal	Min. val. loss	Min. val. loss

in question acquisition, annotation cost, or total training compute. Seed uncertainty does not include variation over independently drawn reduced answer label subsets.

4.6 Decoding, Metrics and Statistical Comparisons

The common decoder uses 384-token question–context windows with 128-token overlap. It considers valid spans with start position no later than end, imposes no additional answer length cap, and returns the literal source text span with the highest summed start/end logit across windows. Exact match (EM) and SQuAD style token F1 lowercase text, remove punctuation and normalise whitespace, retaining articles. This is not identical to the official SQuAD normalisation. Character Frequency overlap is not used as a paper metric. All scores concern the benchmark annotation, not independently assessed legal correctness.

The decoder uses the tokenizer's question/context sequence IDs and offset mapping, considers context tokens only, masks spans with start position greater than end position, and converts the winning token offsets into a literal character slice of the original context. Spans cannot cross a window boundary; overflow windows are mapped back to their source example and the highest raw start plus end logit across windows is retained. No null answer option or additional answer length cap is used.

For Experiment D, the condition comparison is a complete block analysis of variance with training seed as the block. Pairwise confidence intervals use the Student's t distribution over paired seed differences. Holm correction covers the three DAPT versus N contrasts separately for EM and F1. This analysis was specified and locked after validation review but before D test evaluation; it remains a post hoc sensitivity study because D was designed after the earlier A–C results.

Experiment A uses the same blocked analysis and applies Holm correction to all six pairwise contrasts on the full dataset separately for each metric. Experiment C applies Holm correction to the three DAPT versus N contrasts within each fraction of answer labels and metric. The style contrast and the contrast over fractions of answer

labels interaction contrasts are reported with their unadjusted values. These analyses do not constitute one study wide multiplicity correction.

To test whether the DAPT contrast changes with the fraction of answer labels, let $Y_{d,f,s}$ denote test F1 for condition d, label fraction f, and seed s. Experiment C uses the seed-level interaction contrast

$$\Delta_{d,f,s} = (Y_{d,f,s} - Y_{N,f,s}) - (Y_{d,100,s} - Y_{N,100,s}),$$

for $d \in \{A, S, M\}$, $f \in \{25, 50\}$, and $s \in \{7, 42, 99\}$. The mean, confidence interval, and paired test are computed over the three seed-level values of $\Delta_{d,f,s}$. The 100% reference is the three-epoch Experiment A route, not the later Experiment D route with equal update counts. A nonsignificant result is treated as a failure to detect an effect, not as evidence of equivalence. Intervals condition on the fixed test set, corpus selections and training subsets. They do not capture all sources of dataset or source section uncertainty. A clearly labelled post hoc sign flip sensitivity check is reported separately in Section 5.2.3.

5 Results of the Completed Study

Validation summaries are reported separately in Appendix B; they are not substituted for test results. All 62 readers have saved checkpoints and completed evaluations.

5.1 Efficacy and Corpus Composition with Equal Update Counts

Table 7 reports the main comparison with equal update counts on the 344-question test set. Mean F1 is 85.86 for N, 85.85 for A, 85.91 for S, and 85.58 for M. The seed blocked condition test gives $F(3, 12) = 0.474$, $p = 0.706$, and partial $\eta^2 = 0.106$. The corresponding EM test gives $F(3, 12) = 0.245$, $p = 0.863$.

Table 7 Experiment D with equal update counts: test results on 344 questions. EM and F1 use the 0–100 scale; SD is the sample standard deviation across five seeds.

Condition	Mean EM	Mean F1 ± SD	DAPT–N F1 difference (95% interval)
No DAPT	65.76	85.86 ± 1.08	—
Authentic	65.76	85.85 ± 0.46	-0.01 $[-1.34, 1.32]$
Synthetic	65.47	85.91 ± 0.84	$+0.05$ $[-0.80, 0.90]$
Mixed	65.29	85.58 ± 0.75	-0.28 $[-1.16, 0.60]$

All three paired F1 intervals include zero, and all Holm adjusted DAPT versus N p-values equal 1.00. The paired differences vary across seeds. For example, S minus N ranges from -0.70 to $+1.10$ F1 points. Figure 1 shows the per seed scores and paired intervals. The figure displays uncertainty around the observed effects; the zero line

is not an equivalence margin. The intervals also do not rule out effects of practical interest. The widest one, for A minus N, extends from −1.34 to +1.32 F1 points.

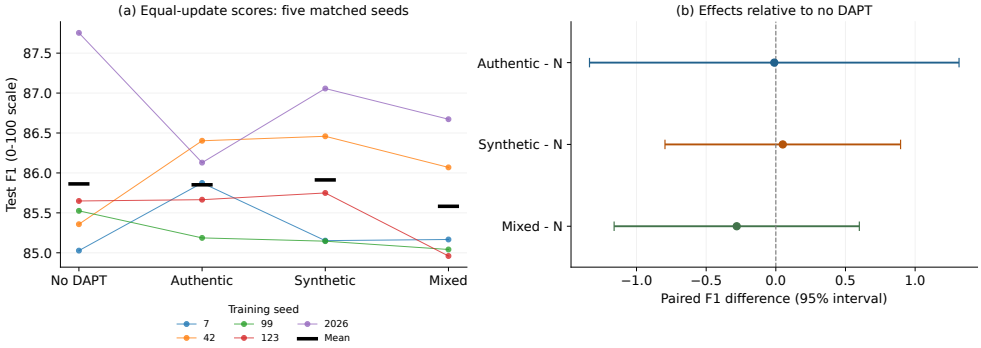


Fig. 1 Experiment D with equal update counts. Left: test F1 for each of five matched training seeds, with condition means marked. Right: paired DAPT minus N F1 differences and 95% Student's t intervals. N uses the matching no DAPT readers from Experiment A because it has no MLM stage.

Answers to RQ1 and RQ2.

Under the schedule with equal update counts, the study does not detect a DAPT benefit on the full dataset or a difference among authentic, synthetic, and mixed adaptation. The absence of a detected effect is not evidence that the routes are equivalent. The result applies to the tested 100-update schedule, fixed corpora, encoder, and downstream route.

5.2 Earlier Three-Epoch Experiments

Table 8 reports the earlier A–C test results. These experiments retain the schedule of three epochs with approximately matched token exposure and its unequal update counts. They provide schedule sensitivity, target alignment, and controls for the fraction of answer labels rather than replacements for Experiment D.

5.2.1 Full Dataset Sensitivity with Matched Token Exposure

Mean F1 is 85.86 for N, 85.75 for A, 85.39 for S and 85.48 for M. The seed blocked test gives $F(3, 12) = 0.635$, $p = 0.607$, and partial $\eta^2 = 0.137$. A, S and M differ from N by -0.12, -0.47 and -0.38 points, with paired 95% intervals $[-1.60, 1.37]$, $[-1.41, 0.47]$ and $[-2.01, 1.24]$. All six F1 contrasts on the full dataset have Holm adjusted $p = 1.00$. The EM condition test is likewise not significant ($F(3, 12) = 0.108$, $p = 0.954$). Figure 2 shows the seed level scores and paired F1 effects.

Experiment A also gives no detected condition effect. Because it uses unequal update counts and ends during warmup, Experiment D provides the cleaner schedule comparison.

Table 8 Earlier three-epoch A–C studies: test results on 344 questions. EM and F1 use the 0–100 scale; SD is the sample standard deviation across seeds.

Experiment	Condition	IPR labels (%)	Seeds	Mean EM	Mean F1 ± SD
A	No DAPT	100	5	65.76	85.86 ± 1.08
A	Authentic	100	5	65.58	85.75 ± 0.59
A	Synthetic	100	5	65.81	85.39 ± 0.67
A	Mixed	100	5	65.35	85.48 ± 0.55
B	Style Only	100	3	65.12	85.60 ± 0.55
C	No DAPT	25	3	62.50	83.54 ± 0.77
C	Authentic	25	3	62.02	83.73 ± 0.34
C	Synthetic	25	3	62.21	83.89 ± 0.65
C	Mixed	25	3	61.43	83.73 ± 0.88
C	No DAPT	50	3	63.28	84.17 ± 0.40
C	Authentic	50	3	63.95	84.35 ± 0.74
C	Synthetic	50	3	63.76	84.60 ± 0.44
C	Mixed	50	3	63.08	84.53 ± 0.64

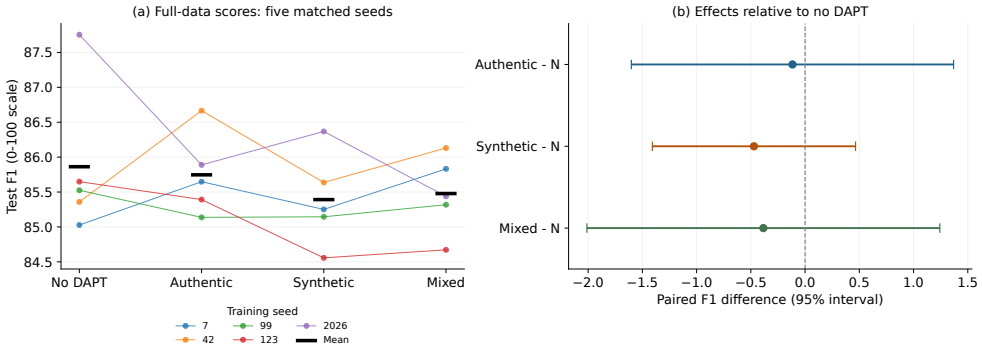


Fig. 2 Three-epoch Experiment A. Left: test F1 for each of five training seeds, with matching seeds connected and condition means marked. Right: paired DAPT minus N F1 differences and 95% Student's t intervals. The zero line denotes no difference, not an equivalence margin.

5.2.2 Target Alignment

Using matching seeds 7, 42 and 99, S reaches 85.34 mean F1 and the untargeted statutory style control reaches 85.60. The difference between S and Style is -0.26 points, with interval $[-1.32, 0.80]$ and uncorrected $p = 0.404$. This comparison uses the S mean across three seeds, not the value across five seeds in the full dataset rows of Table 8.

Answer to RQ3.

The experiment does not detect a target alignment advantage over generic statutory style adaptation at this budget. The control compares particular corpus constructions;

it does not establish that target specific concepts are irrelevant or that generated text is legally accurate.

5.2.3 Fine-Tuning with Fewer Answer Labels

At 25%, the three DAPT versus N F1 intervals include zero. At 50%, a preliminary synthetic versus no DAPT contrast of +0.43 F1 points is observed under the paired-t analysis (interval $[0.30, 0.56]$, within fraction Holm adjusted $p = 0.014$); the exact sign flip sensitivity check gives $p = 0.25$. A and M have positive sample differences of 0.18 and 0.35 points, but their intervals include zero. Figure 3 compares the effects at all three fractions using the same three seeds.

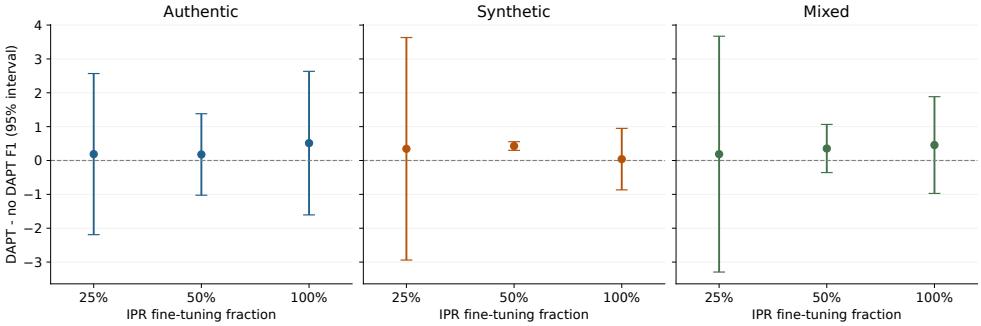


Fig. 3 DAPT minus N test F1 by final supervised fine-tuning fraction. Points are paired mean differences and bars are 95% Student's t intervals. All panels use seeds 7, 42 and 99, including the 100% comparison. The 100% point comes from the three-epoch Experiment A, not Experiment D. Fractions refer to section sampled fine-tuning subsets, not total task resource access. The panels do not establish an interaction with the fraction of answer labels.

The direct interaction comparison is important. For S, $(S - N)_{50} - (S - N)_{100} = 0.39$, with interval $[-0.59, 1.36]$ and unadjusted $p = 0.230$. At 25%, the corresponding difference is 0.31, with interval $[-3.84, 4.45]$, $p = 0.781$. All six A/S/M interaction intervals contain zero (Table 14). A significant contrast at one fraction and a contrast that is not significant at another do not establish different effects.

The three differences between S and N at 50% are 0.463, 0.369 and 0.452 points. The narrow paired-t interval reflects their consistency but has only two degrees of freedom. A post hoc exact two sided sign flip sensitivity check gives $p = 0.25$, with only eight possible sign assignments. This does not replace or invalidate the specified t-test calculation; it illustrates the limited resolution of distribution free inference with three pairs. The contrast is preliminary, warrants replication, and is not a general superiority claim.

Answer to RQ4.

A preliminary +0.43 F1 synthetic versus no DAPT contrast is observed within the fixed fine-tuning with 50% of the answer labels subset under the paired-t analysis. The

exact sign flip sensitivity check does not support that contrast, and the interaction analysis does not establish that DAPT benefits increase as the fraction of answer labels decreases. The design does not estimate total data acquisition savings.

6 Supporting Diagnostics

6.1 Corpus Properties

Table 9 characterises the actual adaptation corpora, rather than only the larger generation pool. Synthetic passages have a narrower length distribution than authentic provisions and a different frequency of drafting markers. The final S corpus covers all seven Act labels in its generation metadata, but metadata coverage does not prove doctrinal coverage or balanced representation.

Table 9 Descriptive properties of the final corpora. Lengths use raw words; token budget matching instead uses truncated tokenizer inputs. Marker frequencies are per 100 words.

Property	A	S	M	Style
Mean words	125.55	130.46	128.17	131.53
Median words	112	130	130	132
10th–90th percentiles	33–229	113–148	56–195	114–148
Type–token ratio	0.0367	0.0331	0.0398	0.0351
Distinct bigram ratio	0.2227	0.2653	0.2788	0.2871
shall	1.323	1.861	1.543	2.015
provided that	0.120	0.335	0.237	0.335
subject to	0.170	0.524	0.335	0.525
Exact duplicate rate	0	0	0	0

A near duplicate diagnostic samples 400 records per corpus and compares all 79,800 within sample pairs using word 5-gram Jaccard similarity at threshold 0.8. It identifies one pair in A and none in S, M or Style; no records are removed on this basis. These checks operate on individual samples; they do not claim that every near duplicate has been excluded. Separate sampled comparisons of authentic text with S and Style each contain 90,000 pairs, with maximum similarities 0.0427 and 0.0448 and no pair above 0.8. This lexical measure does not establish stronger semantic target alignment for S. Token matching also leaves length distributions and drafting style frequencies different. S and Style are close in median length (130 versus 132 words) and in the rates of *provided that* and *subject to*, but they differ in lexical and bigram diversity and in other drafting marker frequencies (Table 9). The style comparison is therefore a control matched by construction, not an intervention with identical distributions.

6.2 Answer Difficulty Diagnostics

Table 10 reports three-epoch Experiment A test results for deterministic answer text groups. Length bins count gold answer words; enumeration uses parenthesised alphanumeric or Roman markers; the other tags detect a semicolon or predefined proviso/exception phrase. The tags are not human annotations of legal difficulty. Groups overlap, and their scores should not be treated as independent experiments.

Table 10 Descriptive Experiment A F1 by gold answer property. Values are means and sample SDs across five seeds; the groups are mechanical and overlapping.

Group	Questions	N	A	S	M
1–10 words	39	80.39 ± 1.22	80.07 ± 1.51	80.47 ± 1.44	80.45 ± 1.96
11–30 words	118	88.69 ± 1.10	89.17 ± 1.07	88.47 ± 1.01	88.73 ± 0.98
31+ words	187	85.22 ± 1.36	84.77 ± 0.38	84.47 ± 0.90	84.47 ± 0.34
Enumerated	70	71.16 ± 3.48	70.32 ± 3.22	69.89 ± 2.80	69.47 ± 2.33
Not enumerated	274	89.62 ± 1.19	89.69 ± 0.31	89.35 ± 0.55	89.57 ± 0.33
Semicolon	41	81.68 ± 1.73	79.82 ± 1.48	78.86 ± 2.44	80.00 ± 1.69
No semicolon	303	86.43 ± 1.04	86.55 ± 0.50	86.27 ± 0.57	86.22 ± 0.50
Proviso/exception	50	85.35 ± 3.99	83.61 ± 2.71	83.52 ± 1.39	83.71 ± 1.91
No tagged phrase	294	85.95 ± 0.73	86.11 ± 0.45	85.71 ± 0.66	85.78 ± 0.33

Enumerated answers have lower mean F1 than not enumerated answers in every condition. DAPT does not show a consistent descriptive advantage in the enumeration, semicolon, or proviso groups. Length alone is not monotonically associated with performance: the 1–10-word group also has relatively low F1. These associations can reflect Act, answer form and question composition; they do not establish that a textual marker causes an error. The analysis provides no practitioner judgement of whether an individual predicted span preserves the provision's legal effect.

6.3 MLM Fit on Validation Data in the Three-Epoch Study

All 19 encoders are evaluated on the same 104 validation contexts and 2,092 masked positions. A fixed 15% masking rule uses the same 80/10/10 mask token, random token and unchanged token allocation, with mask seed 20260829 and maximum length 256. Perplexity is the exponential of masked token loss under this protocol, not an autoregressive likelihood of the statute. The diagnostic is separate from reader checkpoint selection.

Table 11 reports the results. Every adapted condition has lower mean perplexity than the base encoder, with A lowest. Yet A does not improve QA on the full dataset over N, and the overall QA condition test is not significant. Improved masked token prediction is therefore insufficient evidence of improved span extraction under this recipe.

The final validation only values should not be compared numerically with the archived test context diagnostic as if they used identical evaluation inputs and loading procedures. The final diagnostic's in memory bias loading correction is documented in Section A.

Table 11 Experiments A/B: validation MLM perplexity. Means and sample SDs summarise checkpoint level perplexities; lower is better.

Condition	Checkpoints	Perplexity ± SD
No DAPT	1	14.301
Authentic	5	2.938 ± 0.022
Synthetic	5	3.235 ± 0.020
Mixed	5	3.027 ± 0.008
Style Only	3	3.393 ± 0.020

7 Discussion

7.1 Language Fit Is Not Answer Extraction

The Experiment A MLM diagnostic and its QA comparison on the full dataset measure different capabilities. Adaptation improves prediction of masked statutory tokens, whereas answer extraction additionally requires selecting boundaries in response to a question. The observed mismatch does not explain the mechanism of every error, but it shows why MLM perplexity cannot stand in for downstream reader performance.

The benchmark is already approached through an Indian legal encoder and supervised SQuAD transfer. The two analyses on the full dataset that did not detect an effect therefore concern the marginal value of a small additional statutory corpus within that route. It does not compare legal pretraining with no legal pretraining, nor does it establish that DAPT would fail with other encoders, larger corpora, longer training or different optimisation schedules.

Three explanations remain consistent with the evidence: InLegalBERT may already encode much of the useful legal language signal; the MLM objective may improve local token prediction without improving answer boundaries conditioned on the question; or subsequent SQuAD and IndicIPR-QA fine-tuning may attenuate the small adaptation. The present design does not distinguish these mechanisms. The persistent descriptive difficulty of enumerated and qualification bearing answers across conditions suggests that better statutory token prediction alone did not resolve those boundary patterns, but this is not a causal error analysis.

7.2 Implications for Legal Information Access

Gold context evaluation removes the task of finding the supporting provision. A reader's answer is assessed against the benchmark annotation, not against an expert determination of the uniquely governing law. Even exact agreement does not independently validate the source annotation or establish legal completeness. Conversely, token overlap can remain high when an extracted span omits an important exception.

The difficulty diagnostics are useful because they connect aggregate scores with observable answer forms. Their mechanical tags do not measure the legal materiality of an error. A practitioner audit of omitted conditions, added categories and exception boundaries would provide a different form of evidence. The present study supports

separating these evaluations, not treating high gold context F1 as readiness for legal advice.

Generated text is an adaptation resource, not an authority to cite in an answer. Filtering for statutory style neither verifies doctrinal accuracy nor guarantees that rights, obligations or institutional roles are represented correctly. The absence of a detected target alignment advantage also cautions against assuming that benchmark derived concepts necessarily improve extraction.

7.3 What the Controls Establish

The comparison with equal update counts across five seeds is stronger evidence about the marginal adaptation effect than a comparison between the best archived checkpoints. It also addresses the unequal update count and incomplete warmup in Experiment A. Experiments A and D control different parts of exposure and both give no detected condition effect. The other controls do not show a target alignment benefit or an increase in DAPT benefit as the final fraction of answer labels decreases.

Equal update counts are not complete exposure matching. Experiment D processes about 446,000–477,000 nonpadding tokens on average across conditions, and only one corpus selection is used per condition. Corpus provenance, length, lexical distribution, and semantic content also remain linked. A larger study could jointly control updates and token exposure, vary corpus samples, and tune the adaptation duration. Those experiments have not been performed here.

8 Limitations

The central limitation is the short adaptation budget. Experiment D matches 100 update steps, warmup, and learning rate schedule, but it does not match the number of nonpadding tokens processed. Experiment A closely matches the initial corpus token budget, but its update counts and learning rate exposure differ. Neither study matches length distributions or semantic content. No learning rate or adaptation duration search was conducted. The evidence therefore supports a result limited to this recipe, not a broad negative result about statutory DAPT.

Only five seeds are used for each comparison on the full dataset and three for the controls on style and the fraction of answer labels. Experiment D was designed after the A–C results, so it is a post hoc sensitivity analysis rather than an independent confirmatory replication. Its test plan was locked after validation review and before test evaluation. The narrow interval for the 50% synthetic contrast depends on a paired-t analysis with two degrees of freedom. The study on the full dataset did not predefine a smallest effect size or an equivalence margin. Its paired intervals, which extend as far as about ± 1.3 F1 points, cannot exclude modest effects that may matter in practice. Multiplicity correction is within specified families, not across every analysis. The test set, corpus selection and answer label subsets are fixed; the intervals do not jointly estimate corpus, subset, source section and training uncertainty. Fine-tuning with fewer answer labels also retains broader access to text informed by the task and to SQuAD labels.

A prospective confirmation would need more seeds, longer adaptation budgets, at least one additional encoder, and independently resampled adaptation corpora. It should define the smallest practically relevant F1 change before test evaluation. Those additions are needed to distinguish a genuinely small marginal effect from insufficient precision or adaptation exposure.

Provenance safeguards have limits. The raw seed file defect did not propagate according to the generation records, but exact and n-gram overlap checks cannot exclude all semantic overlap or identify data already seen during the base encoder's or generator's original pretraining. The original runner is unavailable for exact historical comparison; the pretest amendment was approved after validation results had been observed. The MLM loader correction occurred after test evaluation. These events are documented rather than described as an untouched preregistered experiment.

The study covers answerable, English, gold context extraction in seven Indian IPR Acts with one encoder family. Questions were machine generated and curated; generator specific patterns may remain. Dataset construction and validation belong to the companion resource paper (Anonymous 2026), not to a new human evaluation in this study. No practitioner assessment of reader errors or generated law like text was conducted. The archived statutory snapshots are not verified as current consolidated law.

Generality across jurisdictions, languages, encoders and independently written questions remains untested. EQUALS concerns Chinese law extraction (Chen et al. 2023); AILQA and DiscoLQA study related legal QA settings (Nigam et al. 2026; Sovrano et al. 2025), but are not direct replications of this English IPR reader protocol. Retrieval, unanswerable questions, multiple provision synthesis and interactive legal research are outside scope.

9 Conclusion

Across five training seeds, the comparison with equal update counts does not detect an extraction gain on the full dataset from authentic, synthetic, or mixed statutory DAPT. Mean test F1 is 85.86 without DAPT, 85.85 with authentic text, 85.91 with synthetic text, and 85.58 with mixed text. The earlier three-epoch comparison reaches the same general result and also does not detect an effect, despite using a different exposure rule. A style control across three seeds does not establish a target alignment advantage. A preliminary 0.43-point synthetic versus no DAPT contrast appears at fine-tuning with 50% of the answer labels under the paired-t analysis, but the exact sign flip check does not support it and the interaction with the fraction of answer labels is not established.

The three-epoch MLM diagnostics improve after adaptation, but this language fit gain does not transfer to detected extraction gains on the full dataset. The findings are bounded by one benchmark, fixed corpus samples, one encoder, and short adaptation schedules. The study supports explicit reporting of source provenance, token and update budgets, checkpoint selection, training seed uncertainty, and answer quality relative to the annotations when evaluating statutory adaptation.

A Analysis Provenance and Reproduction

The analysis record retains the original lock and an approved by authors pretest amendment dated 13 September 2026. At runner review, 44 of 45 originally locked files matched their recorded hashes; the runner differed. The amendment subsequently authorised both the reviewed runner and an evaluator change enforcing amendment and frozen checkpoint verification. The remaining 43 original hashes were unchanged. The current runner reproduces all 88 saved stage scripts at executable code level, but the unavailable original runner prevents an exact comparison with that version. Validation scores had already been observed; final test evaluation followed the amendment. The amendment changed execution and reproducibility infrastructure only; it did not change the stated hypotheses, conditions, evaluation split, metrics, contrast definitions, or multiplicity procedure.

The final MLM diagnostic is a documented corrected posttest diagnostic. The loading correction occurred after test evaluation and did not alter QA predictions, checkpoints, or the frozen QA statistics. Saved checkpoints retain one copy of a shared prediction bias, whereas the loader created a separate zero decoder bias. The corrected loader restores the shared bias in memory while keeping embedding and decoder weight matrices separate. The restored baseline matches the original model's tensors and logits. The saved checkpoint files, training procedure, QA predictions, diagnostic contexts and masks were not changed. The earlier incomplete MLM diagnostic is excluded.

A subsequent reporting audit scores again all 16,168 saved test predictions and 19,176 saved validation predictions from the answer strings, checks their question/reference alignment, and reproduces the reported aggregates. The 47 reader fingerprints match the evaluation records. This verifies consistency of the retained evidence, not independent replication of training. The post hoc sign flip sensitivity in Section 5.2.3 is distinguished from the frozen primary analysis.

Experiment D was designed after the A–C results as a post hoc schedule sensitivity analysis. Its validation results were inspected before the test analysis was frozen. On 2 October 2026, the test lock recorded the protocol, input files, generated scripts, validation results, and 20 reviewed checkpoint fingerprints. Canonical test evaluation then produced 6,880 saved predictions from the 20 condition–seed cells. A separate audit scores again all Experiment D validation and test predictions, checks the 15 new checkpoint fingerprints, verifies the 100-update telemetry, and reproduces the blocked tests and paired contrasts. The five no DAPT cells reuse the matching Experiment A readers. This sequence prevents the Experiment D test results from being presented as preregistered or independent confirmation.

AI assisted manuscript preparation.

An AI assistant (OpenAI Codex) was used to help check consistency between saved evidence and manuscript statements, revise the manuscript, and prepare result driven plotting and verification code. It also executed the documented, frozen Experiment D evaluation and analysis workflow after training had finished. It did not train models, change checkpoints, or alter prediction outputs. Responsibility for the scientific interpretation, verification and final submitted text rests with the authors.

B Validation Results and Interaction Contrasts

Tables 12 and 13 give descriptive validation scores. The same validation set supports supervised checkpoint selection; these are not unseen test estimates or additional evidence of statistical superiority.

Table 12 Experiment D with equal update counts: validation results on 408 questions. EM and F1 use the 0–100 scale; SD is the sample standard deviation across seeds.

Condition	Seeds	Mean EM	Mean F1 ± SD
No DAPT	5	72.60	88.97 ± 1.00
Authentic	5	72.55	88.50 ± 0.87
Synthetic	5	72.89	88.70 ± 0.51
Mixed	5	72.70	88.50 ± 0.80

The validation blocked condition test gives $F(3, 12) = 1.520$, $p = 0.260$ for F1. These values were inspected before the Experiment D test plan was locked and are reported only as validation summaries.

Table 13 Earlier three-epoch A–C studies: validation results on 408 questions. EM and F1 use the 0–100 scale. SD is the sample standard deviation across seeds.

Experiment	Condition	IPR labels (%)	Seeds	Mean EM	Mean F1 ± SD
A	No DAPT	100	5	72.60	88.97 ± 1.00
A	Authentic	100	5	72.99	88.62 ± 0.66
A	Synthetic	100	5	73.04	88.87 ± 0.52
A	Mixed	100	5	72.94	88.73 ± 0.66
B	Style Only	100	3	72.79	88.63 ± 0.61
C	No DAPT	25	3	65.77	84.63 ± 1.22
C	Authentic	25	3	66.09	84.83 ± 1.25
C	Synthetic	25	3	66.34	85.00 ± 1.14
C	Mixed	25	3	66.18	84.73 ± 1.34
C	No DAPT	50	3	68.63	85.90 ± 1.82
C	Authentic	50	3	69.12	85.49 ± 1.46
C	Synthetic	50	3	69.04	85.36 ± 1.25
C	Mixed	50	3	69.20	86.03 ± 0.53

Table 14 reports all six contrasts over fractions of answer labels. Each uses seeds 7, 42 and 99 at both fractions; none of the intervals excludes zero. These values supplement, rather than replace, the within fraction comparisons.

Table 14 Paired F1 interaction contrasts: $\Delta_{d,f,s}$. Intervals use Student's t distribution over three paired seed-level contrasts. p-values are unadjusted.

Fraction	DAPT	Difference	95% interval	p
25%	A	−0.32	[−4.82, 4.17]	0.786
25%	S	+0.31	[−3.84, 4.45]	0.781
25%	M	−0.27	[−4.91, 4.37]	0.827
50%	A	−0.34	[−1.63, 0.96]	0.380
50%	S	+0.39	[−0.59, 1.36]	0.230
50%	M	−0.10	[−0.98, 0.77]	0.665

C Historical Experiments Excluded from Primary Inference

C.1 Checkpoint Inventory

The archived experiments predate the completed study with matched token exposure. Table 15 preserves their selected reader scores; Table 16 records their adaptation corpus composition. These routes differ in corpus scale, provenance and some checkpoint rules. The historical model code A denotes direct InLegalBERT fine-tuning here, not the authentic adaptation condition A in the main experiment.

Table 15 Gold context extractive QA results for selected archived reader routes. E2' is strongest overall, while C' is the strongest DAPT route. Values are percentages on the 344-item test set.

Code	Route	EM	F1
A	InLegalBERT → IPR	63.66	85.02
B'	InLegalBERT → SQuAD → IPR	67.73	86.02
E2'	LEGAL-BERT → SQuAD → IPR	66.86	87.37
H'	Authentic DAPT → SQuAD → IPR	65.99	85.73
C'	Mixed DAPT → SQuAD → IPR	67.44	87.04
I'	Synthetic DAPT → SQuAD → IPR	65.41	85.96

The archived C' route reaches 87.04 F1, compared with 86.02 for B', which uses the same encoder family and SQuAD transfer without DAPT. The paired question bootstrap difference is 1.02 points, with 95% interval [−0.68, 2.84] and Holm adjusted $p = 0.502$. This uncertainty concerns fixed checkpoints and sampled test questions, not retraining. E2' reaches 87.37 with a different encoder family and is not a matched causal control. Three test contexts occur in the archived H'/C' adaptation text. No test QA labels are supplied to MLM, but this context exposure prevents a clean training only interpretation.

Table 16 Composition of the archived DAPT corpora. Parentheses give normalized unique records where applicable. The corpora differ in both size and composition, so H', C', and I' should not be interpreted as a single factor ablation.

Regime	Authentic	Pilot synth.	Main synth.	Total
H' authentic only	3,710 (901)	0	0	3,710 (901)
C' mixed	866 (865)	305	13,043	14,214 (14,213)
I' synthetic only	0	0	13,211	13,211

C.2 Comparison Matched by Record Count

The legacy comparison trains on 901 records per condition with seeds 7, 42 and 99. The token totals are 128,929 for authentic, 146,001 for synthetic and 138,141 for mixed text after the same tokenizer truncation. Synthetic text therefore has 13.2% more tokens than authentic text despite equal record counts. The authentic corpus includes three test contexts associated with six questions; the mixed corpus includes one associated with two questions; the synthetic corpus includes none.

The MLM stage retains its terminal checkpoint. The supervised stages use lowest validation loss selection; earlier blanket descriptions of validation selection at every stage do not describe the executed MLM code. Table 17 reports the archived scores, and Table 18 reports exploratory paired contrasts.

Table 17 DAPT comparison across multiple seeds, matched by record count. Each cell reports EM/F1. The mean and standard deviation columns summarise F1 across the three seeds.

Corpus	Seed 7	Seed 42	Seed 99	Mean F1	SD
auth901	65.70 / 85.66	65.70 / 85.82	67.44 / 86.61	86.03	0.51
synth901	67.73 / 86.49	66.28 / 85.94	66.28 / 84.76	85.73	0.88
mixed901	65.12 / 84.61	66.28 / 85.37	67.15 / 85.96	85.31	0.68

Table 18 Exploratory paired F1 contrasts across seeds for the comparison matched by record count. The reported p-values are unadjusted; the intervals are shown because lack of significance is not evidence that the corpora are equivalent.

Contrast	Mean difference	95% CI	p
auth901 – synth901	0.30	[-3.14, 3.74]	0.745
auth901 – mixed901	0.72	[-0.06, 1.49]	0.057
synth901 – mixed901	0.42	[-3.43, 4.26]	0.686

The seed blocked F1 condition test gives $p = 0.6257$. A one way sensitivity analysis gives $F(2, 6) = 0.7779$, $p = 0.5007$, and sample $\eta^2 = 0.2059$. The observed spread of

means is 0.72 points, compared with pooled within condition SD 0.71. These results do not establish equivalence, and the design does not isolate composition from token exposure or held out context access.

C.3 Archived MLM Diagnostic

Table 19 preserves the archived diagnostic on 113 test contexts, including the three context H'/C' exposure. The diagnostic matched by record count perplexities are 3.01, 3.18 and 3.03 for authentic, synthetic and mixed text, against 11.20 for the baseline. These historical values have different evaluation inputs from the final validation only diagnostic and are not combined with it or used for final model selection.

Table 19 MLM loss and perplexity on 113 test split statutory contexts. This is a secondary diagnostic of statutory language fit, not a fully held out estimate or downstream selection metric. Three contexts occur in the H'/C' DAPT corpora; no test questions or answer labels are used during DAPT, and the exposure is audited in Section 8.

Encoder after MLM stage	Loss	Perplexity	PPL reduction
InLegalBERT baseline	2.4157	11.20	–
Authentic DAPT (H')	0.9131	2.49	77.7%
Mixed DAPT (C')	0.9438	2.57	77.1%
Synthetic DAPT (I')	1.0273	2.79	75.1%

Data and Code Availability

The reported analyses use the fixed IndicIPR-QA splits, adaptation corpus manifests, saved predictions for each question, and experiment and evaluation code. The retained evidence includes the original analysis lock, its dated amendment, checkpoint fingerprints and the MLM loading correction record. Repository and reviewer access information is supplied through the journal submission system. Editors and reviewers can receive the benchmark splits, corpus manifests, frozen predictions, hashes, and analysis code through that system. Subject to dataset and model licences, the final published version will provide a permanent repository link.

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